

Prognosis of juvenile myoclonic epilepsy 45 years after onset

Seizure outcome and predictors



Philine Senf, MD
Bettina Schmitz, PhD
Martin Holtkamp, PhD
Dieter Janz, PhD, FANA

Correspondence to
Dr. Senf:
philine.senf@charite.de

ABSTRACT

Objectives: Juvenile myoclonic epilepsy (JME) is the most common idiopathic generalized epilepsy subsyndrome, contributing to approximately 3% to 11% of adolescent and adult cases of epilepsy. However, little is known about the long-term medical evolution of this clinical entity. The aim of this study was to analyze long-term outcome in a clinically well-defined series of patients with JME for seizure evolution and predictors of seizure outcome.

Methods: In this retrospective cohort study, we analyzed seizure outcome in 66 patients who had JME, were treated at the Department of Neurology, Charité-Universitätsmedizin Berlin, and were initially diagnosed by a single senior epileptologist.

Results: After a mean follow-up time of 44.6 years (20–69 years), 59.1% of patients remained free of seizures for at least 5 years before the last contact. Among the seizure-free patients, 28 (71.8%) were still taking antiepileptic drugs and 11 (28.2%) were off medication for at least the last 5 years. We identified manifestation of additional absence seizures at onset of JME as an independent predictor of an unfavorable outcome regarding seizure freedom.

Conclusions: A significant proportion of patients with JME were seizure-free and off antiepileptic drug therapy in the later course of their disorder. Patients with JME and additional absence seizures might represent a different JME subtype with a worse outcome. *Neurology*® 2013;81:2128–2133

GLOSSARY

AED = antiepileptic drug; **GTCS** = generalized tonic-clonic seizures; **IGE** = idiopathic generalized epilepsy; **JME** = juvenile myoclonic epilepsy.

Juvenile myoclonic epilepsy (JME) is a well-defined subsyndrome of idiopathic generalized epilepsy (IGE) that usually manifests around puberty. The leading symptoms are sudden arrhythmic myoclonic jerks after awakening, often followed or accompanied by generalized tonic-clonic seizures (GTCS). Some patients also exhibit absence seizures.¹ EEG shows generalized spike-wave patterns, typically polyspike waves, and often photosensitivity.^{2,3}

In 80% to 96% of cases, complete seizure control for more than 2 years by use of appropriate anticonvulsants has been described.^{4–6} The best seizure control is still achieved by valproic acid, and only a few studies have examined new antiepileptic drugs (AEDs) for JME treatment.^{7–10} In their original report on JME, published in 1957, Janz and Christian¹ highlighted the rarity of spontaneous remission, as well as the high risk of relapse after AED discontinuation. Further follow-up studies covering periods not longer than 5 years have shown that, in contrast to an initial good response obtained by appropriate treatment, long-term prognosis seems to be rather poor.^{4,11–17} Based on these observations, JME has been described as a “chronic disorder that may require a lifelong therapy”¹² or at least as “rather chronic and long in duration.”⁶ Hence, former therapeutic recommendations suggested lifelong continuation of AED treatment.¹⁸ In contrast, recent large cohorts and a population-based study with follow-up >20 years point to a more favorable course.^{19–22} A recent consensus report recommended AED withdrawal in JME in certain clinical conditions, such as long seizure freedom.²³

From the Department of Psychosomatic Medicine (P.S.), and Epilepsy-Center Berlin-Brandenburg, Department of Neurology (M.H., D.J.), Charité-Universitätsmedizin Berlin; and Center for Epilepsy (B.S.), Department of Neurology, Vivantes-Humboldt Klinikum, Berlin, Germany. Go to Neurology.org for full disclosures. Funding information and disclosures deemed relevant by the authors, if any, are provided at the end of the article.

In this regard, we defined 2 aims for our study: first, we analyzed long-term outcome in a well-defined collective of patients with JME regarding seizure evolution; second, we examined predictors and their negative or positive influence on long-term seizure outcome.

METHODS **Participants.** We included patients who had been diagnosed with JME according to the classification of the International League Against Epilepsy (1989)³ and had been followed up for at least 20 years. Follow-up time was defined as the time between onset of JME (first seizure based on patient records) and last patient contact (personally or via telephone interview between January and July 2011).

The records of the 339 patients with IGE showed that 82 patients had been diagnosed with JME; 190 had absence epilepsy, 51 had GTCS on awakening, and 16 had unclassifiable generalized epilepsy.

Sixteen of 82 patients with a diagnosis of JME did not meet the inclusion criteria, 8 because of insufficient documentation of data and another 8 because of loss to follow-up. All 66 patients finally included had myoclonic jerks and GTCS and 16 patients had additional absence seizures. This seizure type manifested 1 year before myoclonic jerks in one patient, and in the year of first myoclonic seizures or even later in the other 15 patients. At occurrence of absence seizures, all patients were 12 years or older.

All included patients were diagnosed and treated by the same doctor, a senior epileptologist and one of the authors (D.J.).

Procedure and materials. Patient data were analyzed based on the case records and via direct contact by telephone and/or postal letter. Each patient was asked to answer a structured questionnaire regarding seizures and treatment. Forty-six patients answered directly by telephone or by returning the questionnaire. The remaining 20 patients were included based on detailed documentation in their case records, if the follow-up time between their last outpatient visit and the onset of JME was more than 20 years. Because of the retrospective character of the study, during follow-up time, no specific laboratory or EEG investigations were considered in this analysis.

Potential predictors of seizure outcome assessed in this study were age at onset of JME, genetic traits (epilepsy in first-degree relatives), first interictal EEG performed in the hospital (with focus on appearance of spike-wave activity, photosensitivity, focal

abnormalities, or abnormal background activity), appearance of absence seizures, sex, and insurance status (private or state insurance).

Statistical analysis. Data were analyzed using SPSS version 18.0 software (SPSS Inc., Chicago, IL). For comparison of patients with 5-year terminal remission and those without, Fisher exact test was used for categorical dependent variables, and univariate analysis of variance was used for metric-dependent variables. Effect size was calculated using the partial η^2 for metric-dependent variables and the ϕ coefficient for categorical-dependent variables. A Kolmogorov-Smirnov test revealed that all metric variables were normally distributed within the 2 outcome groups. Furthermore, variances were homogeneous between groups, as indicated by Levene test.

Standard protocol approvals, registrations, and patient consents. The study was approved by the Institutional Review Board of Charité–Universitätsmedizin Berlin. Written informed consent was obtained from all participants. The study was neither a public nor a clinical trial.

RESULTS **Study population.** Of 82 patients with a diagnosis of JME, sufficient information on seizure outcome was available for 66 patients (33 female and 33 male), indicating a recruitment rate of more than 80%. Clinical data for all patients included in the study are presented in table 1.

Sixteen patients who were excluded from the study did not differ significantly from the final study population regarding age at epilepsy onset and at investigation, sex, and occurrence of absence seizures (data not shown).

Demographic data of the patients who responded to the questionnaire ($n = 46$) also did not differ significantly from data of patients whose information was retrieved from the records ($n = 20$) regarding age at onset and at investigation, sex, and occurrence of absence seizures (data not shown).

Seizure outcome. After follow-up time of 44.6 ± 13.7 years (range, 20–69 years), 39 of 66 patients with JME (59.1%) had a 5-year terminal remission. Among these 39 seizure-free patients, 28 (71.8%) were still taking AEDs and 11 had been completely off medication (28.2%; or 16.7% of all patients) at least during the last 5 years. The mean terminal seizure-free time of the 39 seizure-free patients was 22.9 ± 10.9 years (range, 5–46 years).

From the remaining 27 patients who were not seizure-free, 26 were treated with AEDs, whereas one woman who had only rare myoclonic jerks was off medication. Of the 27 patients without remission, 4 (14.8%) continued to have only GTCS, 4 (14.8%) had only ongoing myoclonic jerks, and 19 (70.4%) still experienced both types of seizures. None of these patients had ongoing absence seizures. The patients with poor terminal outcome also had seizure-free intervals of up to 30 years (mean, 7.2 ± 8.7 years) during follow-up. Eleven of those 27 patients (40.7%) reported that they had experienced interval seizure-free periods of more than 5 years.

Table 1 Summary of clinical data of all 66 patients included in the study	
Participants with JME (n = 66)	
Age at investigation or last contact, y, mean $\pm$ SD	58.9 $\pm$ 13.8 (min 31, max 84, median 60)
Age at onset, y, mean $\pm$ SD	14.3 $\pm$ 4.1 (min 5, max 33, median 14)
Follow-up time, y, mean $\pm$ SD	44.6 $\pm$ 13.7 (min 20, max 69, median 46)
Sex, n (%)	33 (50) male; 33 (50) female
Absence seizures, n (%)	16 (24.2)
Private insurance, n (%)	40 (60.6)
First interictal EEG, n (%)	33 (50) spike wave; 8 (12.1) diffuse slowing; 6 (9.1) regional slowing; 19 (28.8) normal
Photosensitivity, n (%)	16 (24.2)
Heredity, n (%)	5 (7.6), inherited from first-degree relatives

Abbreviations: JME = juvenile myoclonic epilepsy; max = maximum; min = minimum.

Predictors of seizure outcome. Comparison of clinical variables as potential predictors of seizure outcome in patients with ($n = 39$) and without ($n = 27$) remission demonstrated that additional manifestation of absence seizures was significantly associated with lack of 5-year terminal remission ($p = 0.009$, $\phi = 0.320$).

None of the other clinical variables assessed (sex, insurance type, EEG parameters, photosensitivity, and heredity) were significantly different between groups with and without 5-year terminal remission. All clinical variables of seizure outcome are presented in table 2.

AED treatment. All 66 patients were treated with AEDs at least once after initial diagnosis. Fifty-four patients (81.8%) were still taking AEDs: 38 patients were receiving monotherapy (70.4%) and 16 polytherapy (29.6%). The 2 most frequently used AEDs in mono- or polytherapy were primidone (48.1%) and valproate (51.9%). Newer AEDs were administered only in 3 patients (one in polytherapy), none of whom was seizure-free. All data on AED use and the number of patients with remission for the different AED treatment regimens are summarized in table 3. Primidone showed the most effective treatment results with a 5-year terminal remission rate of 73.3%, compared to 50% with valproate. AED withdrawal was attempted in 14 patients, and only one of those had seizure relapse. Mean age at AED withdrawal was 43.2 ± 9.1 years; preceding seizure-free time was 12.4 ± 10.1 years.

Age at last seizure. Among all patients with remission ($n = 39$), we found a significantly younger age at last seizure in patients who had achieved remission

without AED therapy ($n = 11$) than in patients who had achieved remission and were currently undergoing AED therapy (patients with “controlled epilepsy,” $n = 28$). The mean age of patients with 5-year terminal remission and off all AEDs at the time of their last seizure was 29.3 ± 9.3 years, whereas the mean age of patients with controlled epilepsy at the time of their last seizure was 39.8 ± 15 years ($p = 0.037$). Finally, patients with untreated epilepsy in remission had a trend toward a shorter mean duration of active epilepsy (15.1 ± 10.3 years; median, 13; range, 1–34) compared to those with controlled epilepsy (25.1 ± 14.9 years; median, 23.5; range, 3–52), but the difference was not significant ($p = 0.8$).

DISCUSSION To our knowledge, this long-term study of patients with JME includes the largest cohort and longest follow-up period examining evolution of seizures. Regarding follow-up time and number of patients, there are 3 prior comparable follow-up studies: one study describing the course of 31 patients with a terminal 5-year remission rate of 67.7% (28.6% without AEDs)²¹; one population-based follow-up study of 23 cases (primarily with JME) with a terminal 5- to 23-year remission rate of 30.4% (26.1% without AEDs)¹⁹; and a hospital-based study describing the course of 48 patients until the end of their fourth decade of life, with a 5-year remission rate of 27.1%.²⁰ The mean follow-up period in the present study was 44.6 years, compared with 39.1, 25.8, and 19.6 years, respectively, in these previous long-term studies.

In the current study, the majority of patients exhibited 5-year terminal remission, and almost one-third of those were not receiving AED therapy

Table 2 Clinical variables for patients with and without remission

	Remission ($n = 39$)	No remission ($n = 27$)	p	Effect size
Age at investigation, y, mean $\pm$ SD	59.2 ± 13.4	58.5 ± 14.6	0.830	0.001 ^a
Age at onset, y, mean $\pm$ SD	14.6 ± 3.0	14.0 ± 5.6	0.580	0.005 ^a
Follow-up time, y, mean $\pm$ SD	44.7 ± 13.1	44.5 ± 15.0	0.940	0.001 ^a
Sex, male, n (%)	22 (56.4)	11 (40.7)	0.210	0.159 ^b
Absence seizure, n (%)	5 (12.8)	11 (40.7)	0.009	0.320 ^b
Private insurance, n (%)	22 (56.4)	18 (66.7)	0.410	0.164 ^b
First interictal EEG, n (%)	19 (48.7) spike wave	14 (51.9) spike wave	0.802	0.120 ^b
	6 (15.4) diffuse slowing	2 (7.4) diffuse slowing	0.329	
	9 (23.1) normal	10 (37) normal	0.218	
	5 (12.8) asymmetry/focal abnormality	1 (3.7) asymmetry/focal abnormality	0.205	
Photosensitivity, n (%)	12 (30.8)	4 (14.8)	0.140	0.183 ^b
Heredity, n (%)	4 (10.3)	1 (3.7)	0.320	0.122 ^b

^aPartial η^2 .

^bThe ϕ coefficient.

Table 3 Overview of AED treatment and remission in all patients with JME included in this study

	AED therapy, n (%)	5-y terminal remission
Monotherapy		
Valproate	18 (47.4)	9 yes, 9 no
Primidone	15 (39.5)	11 yes, 4 no
Phenobarbital	1	Yes
Phenytoin	1	Yes
Lamotrigine	1	No
Carbamazepine	1	No
Topiramate	1	No
Polytherapy		
VPA + primidone	8 (50)	4 yes, 4 no
Primidone + phenytoin	2	Yes
Ethosuximide + sultiam	1	Yes
CBZ + VPA	1	No
VPA + phenytoin	1	No
Phenobarbital + clobazam	1	No
Primidone + LEV	1	No
Phenytoin + phenobarbital	1	No

Abbreviations: AED = antiepileptic drug; CBZ = carbamazepine; JME = juvenile myoclonic epilepsy; LEV = levetiracetam; VPA = valproate.

for that time period. Therefore, 11 of 66 patients with JME were in remission and off AEDs for more than the last 5 years. Thus, our data support the results of the studies mentioned above observing a generally better prognosis of JME than previously thought. This contradicts the long-standing leading opinion that lifelong treatment of JME is required to maintain seizure freedom. Thus, our results are in line with other recent studies and may provide a new and more optimistic perspective on the long-term outcome of JME.^{19,21}

The second most outstanding finding of our study was the significantly higher incidence of absence seizures in the group of patients without remission. Considering this information, the manifestation of absence seizures in JME seems to be a negative predictor of seizure outcome. This result is consistent with prior findings of a lower remission rate among patients with childhood absence epilepsy evolving to JME than among patients with “classic” JME, JME with adolescent absence, or JME with astatic seizures.⁶ However, follow-up time in that study varied from 1 to 41 years; therefore, interpretation of these data regarding long-term aspects seems to be difficult. In this particular IGE subgroup of patients with multiple seizure types including absences, poor response to AED therapy has been reported.²⁴ Accordingly, other data suggest that patients with JME with only myoclonic jerks have a better long-term

prognosis than patients with additional GTCS and absence seizures.^{24,25} A recent population-based study excluded patients with childhood absence epilepsy evolving into JME from their follow-up study because this constellation might display a separate subtype of JME with different outcome.¹⁹ They reported that 60% of included patients had a history suggestive of at least some absence seizures during the course of JME (only patients with unrecognized absence seizures before diagnosis of JME were permitted). If this was true, one may speculate that the high rate of additional absence seizures (60%) caused the relatively low remission rate (30.4%) reported in their study.

Our findings support the hypothesis that JME may be a heterogeneous syndrome incorporating different subtypes characterized by different outcomes. They also argue that there is a specific subtype of JME with additional absence seizures with a less favorable outcome as suggested previously.²⁶

We did not find any significant associations for other clinical variables of seizure outcome that were examined (table 2). Consistent with prior findings, we did not observe a significant association between photosensitivity and long-term seizure outcome.^{20,21} Moreover, we did not find any association between the terminal remission rate and specific EEG findings or family history of epilepsy.²⁷

A great proportion (81.8%) of our sample was still taking AEDs and 40% of those patients still had seizures at the last contact. However, it is possible that many patients had adjusted such that they tolerate rare occurrence of seizures rather than follow a strict regimen of AED intake. We did not identify any case with spontaneous healing such as the case in the original description of JME by Janz and Christian.¹ They reported a case of spontaneous remission of seizures without any AED treatment for at least 15 years after a 4-year period with occurrence of myoclonic jerks. An interesting finding of the present study was that primidone seems to be highly effective in JME, which may be considered in patients who do not respond to valproate and the modern compounds used for this indication or in female patients with high teratogenic risk. We hypothesize that in addition to its direct antiepileptic effect, primidone and in particular its active metabolite phenobarbital possibly improve disturbed sleep, eventually resulting in decreased seizure risk the next day.

Another remarkable finding of this study was that patients with JME with seizure freedom but without AED therapy for more than 5 terminal years were significantly younger at the time of their last seizure than patients who had seizure freedom but were still receiving AED treatment. This is consistent with the finding that in some JME cases, there is a relatively short active phase followed by long-term remission without AEDs.¹⁹ Furthermore, the current results are in line

with previous data demonstrating that a shorter duration to achieving seizure freedom was significantly associated with a favorable long-term outcome.²¹ This suggests that excellent outcome in JME is more likely at a younger age, probably before the age of 40 years. However, it should be noted that even in patients with a relatively short active phase of JME, this still lasted approximately 15 years.

This study has strengths and limitations that need to be considered. A unique feature of this “single-doctor study” is that all patients were treated by only one specialized epileptologist for decades. Continuous treatment by one physician may explain the high enrollment rate of more than 80% many decades after diagnosis. Furthermore, we can assume homogeneous diagnostic criteria and treatment strategies, with presumably less cases of misdiagnosis and inappropriate treatment. Many of our patients made a special effort to see a specialized epileptologist, which implies high motivation and probably a higher educational and socioeconomic background. The outcome of our patients may have been positively influenced by an optimal treatment setting, including regularity of visits and repetitive

discussion of methods to improve AED compliance and identification of seizure-provoking behavior. These treatment issues may be of particular importance for patients with JME, many of whom have neuropsychological impairments that can be relevant regarding treatment adherence.²⁸ Thus, our sample may not be representative for JME patients in general. There could be a selection bias in our sample toward more compliant and motivated patients, resulting in an overestimation of positive outcomes.

This study analyzed the long-term outcome in a well-defined sample of patients with JME. Seizure remission occurred in 59.1% of patients, and 16.7% of all patients remained seizure-free without AEDs for at least the terminal 5 years. Thus, seizures could be controlled in the majority of cases, and a significant proportion of patients in addition were off AED therapy. Therefore, our study supports the results of previous work showing that lifelong treatment in patients with JME may no longer apply as a general prognostic guideline. We identified occurrence of absence seizures at the onset of JME as a significant predictor for poor outcome. This finding supports the assumption that patients with JME and additional absence seizures might represent a different JME subtype with a worse outcome. Furthermore, we found that complete seizure freedom without AED therapy is more likely when seizures subside at a younger age, probably before the age of 40 years.

Comment: A reappraisal of juvenile myoclonic epilepsy

The education of neurologists on the syndrome of juvenile myoclonic epilepsy (JME) has been one of the major successes in the treatment of epilepsy. For a long time, the original author to whom this condition is eponymously linked, as well as other experts, conceived of JME as a lifelong condition in which seizure relapse upon drug discontinuation is invariable. This report by Senf et al. on a group of 66 patients with JME followed for a mean of 44.6 years gives a reappraisal of several aspects of the condition.¹

Treatment response and long-term remission are more favorable than previously thought possible. Fifty-nine percent of patients attained 5-year terminal remission, with 16.6% of the cohort no longer taking antiepileptic medication. Other recent long-term outcome studies on JME report even slightly higher chances of being seizure free off medications (19%–26% of patients followed for 20 years or longer).^{2,3} Even in those not completely seizure free, there were only sporadic seizures with many years of seizure freedom in between.

The other face of JME is that a proportion remains resistant to treatment despite the advent of more and newer drugs. Predictors of seizure persistence have been hard to come by. Surprisingly, no consistent EEG variables can be identified. This study found the coexistence of absence seizures to be a predictor of disease persistence. When dealing with refractory JME, it may be well to remember the possible contribution of older therapies such as primidone, acetazolamide, and clonazepam. This study found primidone to be particularly efficacious when compared to valproate. Observation bias is possible, as this cohort was assembled at a time when only older drugs were available, and those who were responders then stayed on the same treatment.

1. Senf P, Schmitz B, Holtkamp M, Janz D. Prognosis of juvenile myoclonic epilepsy 45 years after onset: seizure outcome and predictors. *Neurology* 2013;81:2128–2133.
2. Camfield CS, Camfield PR. Juvenile myoclonic epilepsy 25 years after seizure onset. *Neurology* 2009;73:1041–1045.
3. Geithner J, Schneider F, Wang Z, et al. Predictors for long-term seizure outcome in juvenile myoclonic epilepsy: 25–63 years of follow-up. *Epilepsia* 2012;53:1379–1386.

Norman K. So, MB, BChir

From the Epilepsy Center, Cleveland Clinic, Cleveland, OH.

Study funding: No targeted funding reported.

Disclosure: N. So reports no disclosures relevant to the manuscript. Go to Neurology.org for full disclosures.

AUTHOR CONTRIBUTIONS

Philine Senf: wrote the manuscript, designed the experiments, and performed the statistical analysis. Bettina Schmitz: designed the experiments. Martin Holtkamp: designed the experiments and performed the statistical analysis. Dieter Janz: diagnosed and managed the patients.

STUDY FUNDING

No targeted funding reported.

DISCLOSURE

P. Senf reports no disclosures. B. Schmitz has served on scientific advisory boards for UCB, Novartis, GlaxoSmithKline, Pfizer Inc., Eisai Inc., Desitin Pharmaceuticals, GmbH, and Janssen. She has received reimbursement for traveling expenses and/or speaker honoraria from UCB, Eisai Inc., and GlaxoSmithKline; she serves on the speakers bureaus of GlaxoSmithKline, Pfizer Inc., Janssen, Desitin Pharmaceuticals, GmbH, Eisai Inc., Novartis, and Sanofi-Aventis, and on the editorial boards of *Epilepsy and Behaviour* and *Zeitschrift für Epileptologie*. She receives royalties for publication in *The Neuropsychiatry of Epilepsy* (Cambridge University Press, 2002), *Paroxysmale Störungen in der Neurologie* (Springer, 2005), *Cognitive Dysfunction in Children with Temporal Lobe Epilepsy* (John Libbey, 2005), *Seizures, Affective Disorders and Antiepileptic Drugs* (Clarius Press Ltd., 2002), *Psychiatrische Epileptologie* (Georg Thieme, 2005), *Epilepsien: Taschenatlas Spezial* (Georg Thieme, 2005), *Forced Normalization* (Wrightson Biomedical Publishing, 1998), and *Juvenile Myoclonic Epilepsy: The Janz Syndrome* (Wrightson Biomedical Publishing, 2000). She has received research support from UCB, Sanofi-Aventis, Novartis, Pfizer Inc., Janssen, Desitin Pharmaceuticals, GmbH, Eisai Inc., GlaxoSmithKline, DFG, and Deutsche Gesellschaft für Epileptologie. M. Holtkamp has served on scientific advisory boards

for UCB, Eisai Inc., and Janssen. He has received reimbursement for traveling expenses and/or speaker honoraria from UCB, Eisai Inc., Desitin, Janssen, GlaxoSmithKline, and ViroPharma. D. Janz reports no disclosures. Go to Neurology.org for full disclosures.

Received June 14, 2013. Accepted in final form September 13, 2013.

REFERENCES

1. Janz D, Christian W. Impulsiv - petit mal. *J Neurol* 1957; 176:346–386. Translated into English by Genton P. In: Malatosses A, Genton P, Hirsch E. Idiopathic Generalized Epilepsies: Clinical, Experimental and Genetic Aspects. London: John Libbey; 1994:229–251.
2. Thomas P, Genton P, Gélinde P, Wolf P. Juvenile myoclonic epilepsy. In: Roger J, Bureau M, Dravet CH, Genton P, Tassinari CA, Wolf P, editors. *Epileptic Syndromes in Infancy, Childhood and Adolescence*, 4th ed. London: John Libbey; 2005:367–388.
3. Commission on Classification and Terminology of the International League Against Epilepsy (ILAE). Proposal for revised classification of epilepsies and epileptic syndromes. *Epilepsia* 1989;30:389–399.
4. Delgado-Escueta AV, Enrile-Bacsal F. Juvenile myoclonic epilepsy of Janz. *Neurology* 1984;34:285–294.
5. Christe W, Janz D. Valproate in idiopathic generalised epilepsies. *Boll Lega Ital Epil* 1988;61:11–15.
6. Martínez-Juárez IR, Alonso ME, Medina MT, et al. Juvenile myoclonic epilepsy subsyndromes: family studies and long-term follow-up. *Brain* 2006;129:1269–1280.
7. Noachtar S, Andermann E, Meyvisch P, et al. Levetiracetam for the treatment of idiopathic generalized epilepsy with myoclonic seizures. *Neurology* 2008;70:607–616.
8. Morris GL, Hammer AE, Kustra RP, Messenheimer JA. Lamotrigine for patients with juvenile myoclonic epilepsy following prior treatment with valproate: results of an open-label study. *Epilepsy Behav* 2004;5:509–512.
9. Kothare SV, Valencia I, Khurana DS, Hardison H, Melvin JJ, Legido A. Efficacy and tolerability of zonisamide in juvenile myoclonic epilepsy. *Epileptic Disord* 2004;6:267–270.
10. Biton V, Bourgeois BF; YTC/YTCE Study Investigators. Topiramate in patients with juvenile myoclonic epilepsy. *Arch Neurol* 2005;62:1705–1708.
11. Janz D, Kern A, Mössinger HJ, Puhlmann U. Relapse prognosis following reduction of drugs in epilepsy treatment [in German]. *Nervenarzt* 1983;54:525–529.
12. Penry JK, Dean JC, Riela AR. Juvenile myoclonic epilepsy: long-term response to therapy. *Epilepsia* 1989;30:S19–S23.
13. Shinnar S, Berg AT, Moshé SL, et al. Discontinuing anti-epileptic drugs in children with epilepsy: a prospective study. *Ann Neurol* 1994;35:534–545.
14. Panayiotopoulos CP, Obeid T, Tahan AR. JME: a 5-year prospective study. *Epilepsia* 1994;35:285–296.
15. Atakli D, Sözüer D, Atay T, Baybas S, Arpacı B. Misdiagnosis and treatment in juvenile myoclonic epilepsy. *Seizure* 1998;7:63–66.
16. Sokic D, Ristic AJ, Vojvodic N, Jankovic S, Sindjelic AR. Frequency, causes and phenomenology of late seizure recurrence in patients with juvenile myoclonic epilepsy after a long period of remission. *Seizure* 2007;16:533–537.
17. Guarana MS, Filho GM, Lin K, Guilhoto LM, Caboclo LO, Yacubian EM. Prognosis of juvenile myoclonic epilepsy is related to endophenotypes. *Seizure* 2011;20:42–48.
18. Calleja S, Salas-Puig J, Ribacoba R, Lahoz CH. Evolution of juvenile myoclonic epilepsy treated from the outset with sodium valproate. *Seizure* 2001;10:424–427.
19. Camfield CS, Camfield PR. Juvenile myoclonic epilepsy 25 years after seizure onset: a population based study. *Neurology* 2009;73:1041–1045.
20. Baykan B, Altındag EA, Bebek N, et al. Myoclonic seizures subside in the fourth decade in juvenile myoclonic epilepsy. *Neurology* 2008;70:2123–2129.
21. Geithner J, Schneider F, Wang Z, et al. Predictors for long-term seizure outcome in juvenile myoclonic epilepsy: 25–63 years of follow-up. *Epilepsia* 2012;53:1379–1386.
22. Baykan B, Martínez-Juárez IE, Altındag EA, Camfield CS, Camfield PR. Lifetime prognosis of juvenile myoclonic epilepsy. *Epilepsy Behav* 2013;28(suppl 1):S18–S24.
23. Kasteleijn-Nolst Trenité DG, Schmitz B, Janz D, et al. Consensus on diagnosis and management of JME: from founder's observations to current trends. *Epilepsy Behav* 2013;28(suppl 1):S87–S90.
24. Gelisse P, Genton P, Thomas P, Rey M, Samuelian JC, Dravet C. Clinical factors of drug resistance in juvenile myoclonic epilepsy. *J Neurol Neurosurg Psychiatry* 2001;70:240–243.
25. Jain S, Padma MV, Maheshwari MC. Occurrence of only myoclonic jerks in juvenile myoclonic epilepsy. *Acta Neurol Scand* 1997;95:263–267.
26. Panayiotopoulos CP, Obeid T, Waeed G. Differentiation of typical absence seizures in epileptic syndromes: a video EEG study of 224 seizures in 20 patients. *Brain* 1989;112: 1039–1056.
27. Jayalakshmi SS, Srinavasa Rao B, Sailaja S. Focal clinical and electroencephalographic features in patients with juvenile myoclonic epilepsy. *Acta Neurol Scand* 2010;122:115–123.
28. Gelisse P, Thomas P, Samuelian JC, Gentin P. Psychiatric disorders in juvenile myoclonic epilepsy. *Epilepsia* 2007; 48:1032–1033.